Venn Diagrams & Conditional Probability

Extra practice: 13 exam-style questions with worked answers

Key results to know

- The four regions of a Venn diagram: $P(A \cap B')$ (A only), $P(A \cap B)$ (both), $P(A' \cap B)$ (B only), $P(A' \cap B')$ (neither). They add up to 1.
- $P(A) = P(A \cap B) + P(A \cap B')$ and $P(B) = P(A \cap B) + P(A' \cap B)$
- $P(A \cup B) = P(A) + P(B) - P(A \cap B) = P(A) + P(A' \cap B)$
- $P(B \mid A) = \dfrac{P(A \cap B)}{P(A)}$, so $P(A \cap B) = P(B \mid A)\,P(A)$
- $P(B) = P(B \mid A)\,P(A) + P(B \mid A')\,P(A')$ (the two branches of a tree that end in B)
- A and B are **independent** if and only if $P(A \cap B) = P(A)\,P(B)$.

Method tip: always fill in the Venn diagram region by region *starting from the overlap*. To prove "not independent", calculate both $P(A \cap B)$ and $P(A)P(B)$, then state clearly that they are not equal.

Section A – Venn diagrams and independence

1. **[Maximum mark: 6]**

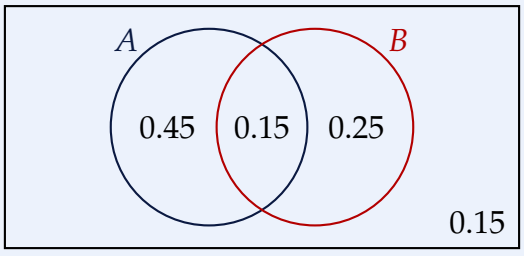
Consider two events, A and B, such that $P(A) = 0.6$, $P(A \cap B') = 0.45$ and $P(A' \cap B) = 0.25$.

(a) By drawing a Venn diagram, or otherwise, find $P(A \cup B)$. [3]

(b) Show that the events A and B are not independent. [3]

Answer

(a) Start with the overlap: $P(A \cap B) = P(A) - P(A \cap B') = 0.6 - 0.45 = 0.15$.
Neither $= 1 - (0.45 + 0.15 + 0.25) = 0.15$.



$$P(A \cup B) = 0.45 + 0.15 + 0.25 = \boxed{0.85}$$

(b) $P(B) = 0.15 + 0.25 = 0.4$, so $P(A)\,P(B) = 0.6 \times 0.4 = 0.24$.

But $P(A \cap B) = 0.15 \neq 0.24$, so A and B are **not independent**.

2. **[Maximum mark: 7]**

Consider two events, A and B, such that $P(A) = 0.3$, $P(B) = 0.5$ and $P(A \cup B) = 0.65$.

(a) Find $P(A \cap B)$. [2]

(b) Show that the events A and B are independent. [2]

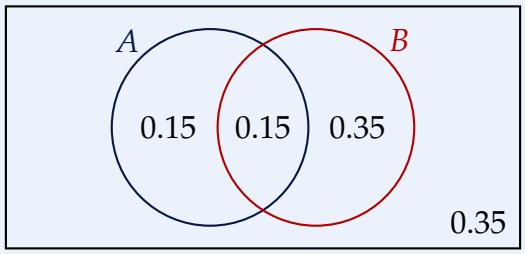
(c) By drawing a Venn diagram, or otherwise, find

(i) $P(A' \cap B')$;

(ii) $P(A \mid B')$. [3]

Answer

(a) $P(A \cap B) = P(A) + P(B) - P(A \cup B) = 0.3 + 0.5 - 0.65 = \boxed{0.15}$
(b) $P(A)\,P(B) = 0.3 \times 0.5 = 0.15 = P(A \cap B)$, so A and B are **independent**.
(c) A only $= 0.3 - 0.15 = 0.15$, B only $= 0.5 - 0.15 = 0.35$, neither $= 1 - 0.65 = 0.35$.



(i) $P(A' \cap B') = \boxed{0.35}$ (the region outside both circles)
(ii) $P(B') = 1 - 0.5 = 0.5$. The part of B' that is inside A is "A only" $= 0.15$:

$$P(A \mid B') = \frac{P(A \cap B')}{P(B')} = \frac{0.15}{0.5} = \boxed{0.3}$$

This equals $P(A)$, which is what we expect when A and B are independent.

3. **[Maximum mark: 7]**

Events A and B are independent, with $P(A) = 0.4$ and $P(A \cup B) = 0.7$.

(a) Show that $P(B) = 0.5$. [3]

(b) By drawing a Venn diagram, or otherwise, find the probability that exactly one of the events A or B occurs. [3]

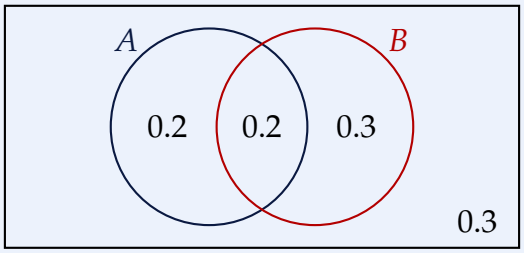
(c) Write down $P(A \mid B)$. [1]

Answer

(a) Let $P(B) = p$. Since A and B are independent, $P(A \cap B) = 0.4p$. Then

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$
$$0.7 = 0.4 + p - 0.4p$$
$$0.3 = 0.6p \Rightarrow p = 0.5 \quad \blacksquare$$

(b) $P(A \cap B) = 0.4 \times 0.5 = 0.2$. A only $= 0.4 - 0.2 = 0.2$, B only $= 0.5 - 0.2 = 0.3$, neither $= 0.3$.



$$P(\text{exactly one}) = 0.2 + 0.3 = \boxed{0.5}$$

(c) A and B are independent, so $P(A \mid B) = P(A) = \boxed{0.4}$.

Section B – Proofs and conditional probability

4. **[Maximum mark: 9]**

Consider two events A and B defined in the same sample space.

(a) Show that $P(A \cup B) = P(B) + P(A \cap B')$. [3]

Given that $P(A \cup B) = \frac{3}{5}$, $P(A \mid B) = \frac{1}{3}$ and $P(A \mid B') = \frac{1}{2}$,

(b) (i) show that $P(B) = \frac{1}{5}$;

(ii) hence find $P(A)$. [6]

Answer

(a) $P(A \cup B) = P(A) + P(B) - P(A \cap B)$.
Also $A = (A \cap B) \cup (A \cap B')$, and these are mutually exclusive, so $P(A) = P(A \cap B) + P(A \cap B')$.
Substituting: $P(A \cup B) = P(A \cap B) + P(A \cap B') + P(B) - P(A \cap B) = P(B) + P(A \cap B')$. $\blacksquare$

(b)(i) Let $P(B) = y$. Then $P(A \cap B') = P(A \mid B')\,P(B') = \frac{1}{2}(1 - y)$. Using part (a):

$$y + \frac{1-y}{2} = \frac{3}{5} \Rightarrow 2y + 1 - y = \frac{6}{5} \Rightarrow y = \frac{1}{5} \quad \blacksquare$$

(b)(ii)

$$P(A) = P(A \mid B)P(B) + P(A \mid B')P(B') = \frac{1}{3} \cdot \frac{1}{5} + \frac{1}{2} \cdot \frac{4}{5} = \frac{1}{15} + \frac{6}{15} = \boxed{\frac{7}{15}}$$

5. **[Maximum mark: 8]**

Consider two events A and B defined in the same sample space.

(a) Show that $P(A \cap B') = P(A) - P(A \cap B)$. [2]

Given that $P(A) = \frac{2}{5}$, $P(B \mid A) = \frac{1}{4}$ and $P(A \cup B) = \frac{7}{10}$,

(b) find $P(A \cap B)$; [1]

(c) show that $P(B) = \frac{2}{5}$; [2]

(d) determine, with a reason, whether A and B are independent; [2]

(e) find $P(A' \cap B')$. [1]

Answer

(a) $A = (A \cap B) \cup (A \cap B')$, and these two events are mutually exclusive, so

$$P(A) = P(A \cap B) + P(A \cap B') \;\Rightarrow\; P(A \cap B') = P(A) - P(A \cap B) \qquad \blacksquare$$

(b) $P(A \cap B) = P(B \mid A)\, P(A) = \frac{1}{4} \times \frac{2}{5} = \boxed{\frac{1}{10}}$

(c) $P(B) = P(A \cup B) - P(A) + P(A \cap B) = \frac{7}{10} - \frac{4}{10} + \frac{1}{10} = \frac{4}{10} = \frac{2}{5}$ $\blacksquare$

(d) $P(A)\, P(B) = \frac{2}{5} \times \frac{2}{5} = \frac{4}{25}$, but $P(A \cap B) = \frac{1}{10} \neq \frac{4}{25}$, so A and B are **not independent**.

(Or: $P(B \mid A) = \frac{1}{4} \neq P(B) = \frac{2}{5}$.)

(e) $P(A' \cap B') = 1 - P(A \cup B) = 1 - \frac{7}{10} = \boxed{\frac{3}{10}}$

6. **[Maximum mark: 8]**

For two events A and B, it is known that $P(B \mid A) = \frac{2}{3}$, $P(B \mid A') = \frac{1}{4}$ and $P(B) = \frac{5}{12}$.

(a) Draw a tree diagram to represent this information, using $P(A) = x$. [2]

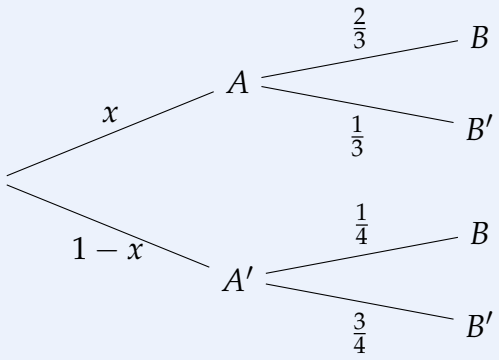
(b) Show that $P(A) = \frac{2}{5}$. [2]

(c) Find $P(A \cup B)$. [2]

(d) Find $P(A \mid B)$. [2]

Answer

(a)



(b) Add the two branches that end in B:

$$\frac{2}{3}x + \frac{1}{4}(1-x) = \frac{5}{12} \;\Rightarrow\; 8x + 3 - 3x = 5 \;\Rightarrow\; 5x = 2 \;\Rightarrow\; x = \frac{2}{5} \qquad \blacksquare$$

(c) $P(A \cup B) = P(A) + P(A' \cap B) = \frac{2}{5} + \frac{3}{5} \times \frac{1}{4} = \frac{8}{20} + \frac{3}{20} = \boxed{\frac{11}{20}}$

(d) $P(A \cap B) = \frac{2}{3} \times \frac{2}{5} = \frac{4}{15}$, so

$$P(A \mid B) = \frac{P(A \cap B)}{P(B)} = \frac{4/15}{5/12} = \frac{4}{15} \times \frac{12}{5} = \boxed{\frac{16}{25}}$$

Section C – School paper questions

Tree diagrams: without replacement

When items are taken *without replacement*, the second-stage probabilities change: there is one fewer item in total, and one fewer of the colour already taken. Each pair of branches from the same point must add to 1. Multiply *along* a path, and add *between* paths.

7. **[Maximum mark: 15]**

A bag contains 4 red and 6 blue marbles. Two marbles are drawn at random, successively and without replacement.

(a) Copy and complete a tree diagram to show all the probabilities for the two draws. [3]

(b) Find the probability that:

(i) both marbles are the same colour; [3]

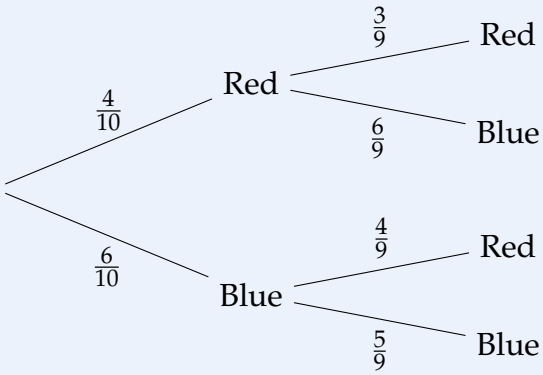
(ii) at least one red marble is drawn; [3]

(iii) the first marble drawn is red, given that at least one red marble is drawn; [3]

(iv) both marbles drawn are red, given that at least one red marble is drawn. [3]

Answer

(a) After a red is taken, 3 red and 6 blue remain (9 in total). After a blue is taken, 4 red and 5 blue remain.



Path probabilities: RR $= \frac{12}{90}$, RB $= \frac{24}{90}$, BR $= \frac{24}{90}$, BB $= \frac{30}{90}$ (total $\frac{90}{90} = 1$ ✓).

(b)(i) $P(\text{same}) = P(RR) + P(BB) = \frac{12}{90} + \frac{30}{90} = \frac{42}{90} = \boxed{\frac{7}{15}}$

(ii) $P(\text{at least one red}) = 1 - P(BB) = 1 - \frac{30}{90} = \frac{60}{90} = \boxed{\frac{2}{3}}$

(iii) "First is red" = RR or RB $= \frac{36}{90}$. This is already inside "at least one red", so

$$P(\text{1st red} \mid \text{at least one red}) = \frac{36/90}{60/90} = \frac{36}{60} = \boxed{\frac{3}{5}}$$

(iv) $P(RR \mid \text{at least one red}) = \dfrac{12/90}{60/90} = \dfrac{12}{60} = \boxed{\dfrac{1}{5}}$

8. **[Maximum mark: 6]**

In a class of students, 35% play football, 25% play basketball, and 50% play neither sport.

(a) Copy and complete a Venn diagram to represent this information. [3]

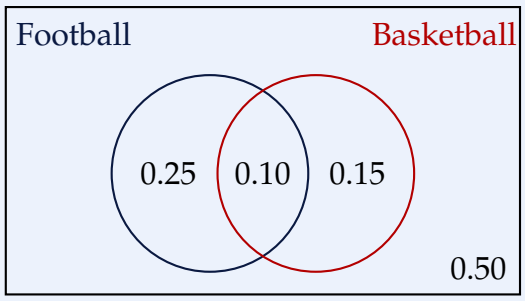
(b) Find the probability that a randomly selected student:

(i) plays both football and basketball; [1]

(ii) plays basketball, given that they play football. [2]

Answer

(a) $P(F \cup B) = 1 - 0.5 = 0.5$, so $P(F \cap B) = 0.35 + 0.25 - 0.5 = 0.10$.
Football only $= 0.35 - 0.10 = 0.25$, basketball only $= 0.25 - 0.10 = 0.15$.



(b)(i) $P(F \cap B) = \boxed{0.1}$

(ii) $P(B \mid F) = \dfrac{P(F \cap B)}{P(F)} = \dfrac{0.10}{0.35} = \boxed{\dfrac{2}{7} \approx 0.286}$

9. **[Maximum mark: 4]**

Let A and B be events such that $P(A) = 0.6$, $P(B) = 0.3$ and $P(A \cup B) = 0.7$. Find $P(A \mid B)$.

Answer

$P(A \cap B) = P(A) + P(B) - P(A \cup B) = 0.6 + 0.3 - 0.7 = 0.2$

$$P(A \mid B) = \frac{P(A \cap B)}{P(B)} = \frac{0.2}{0.3} = \boxed{\frac{2}{3} \approx 0.667}$$

10. **[Maximum mark: 5]**

(a) A biased coin has probability 0.3 of landing on heads. The coin is tossed 8 times. Find the expected number of heads. [2]

(b) Events A and B are independent, with $P(A) = 0.4$ and $P(B) = 0.25$. Find (i) $P(A \cap B)$; (ii) $P(A \cup B)$. [3]

Answer

(a) $X \sim B(8,\ 0.3)$, so $E(X) = np = 8 \times 0.3 = \boxed{2.4}$

(b)(i) Independent, so $P(A \cap B) = P(A)\,P(B) = 0.4 \times 0.25 = \boxed{0.1}$

(ii) $P(A \cup B) = 0.4 + 0.25 - 0.1 = \boxed{0.55}$

11. **[Maximum mark: 6]**

A bag contains 2 green and 5 red balls. Two balls are taken out successively, without replacement.

(a) Copy and complete a tree diagram to show the information given. [2]

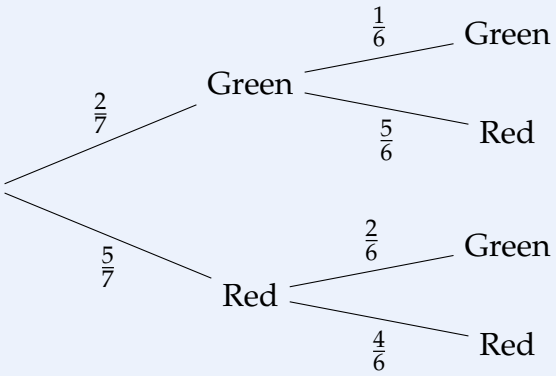
(b) Find the probability that:

(i) at least one green ball is chosen; [2]

(ii) red is taken on the first pick, given that at least one green ball is chosen. [2]

Answer

(a)



Path probabilities: GG $= \frac{2}{42}$, GR $= \frac{10}{42}$, RG $= \frac{10}{42}$, RR $= \frac{20}{42}$.

(b)(i) $P(\text{at least one green}) = 1 - P(RR) = 1 - \frac{20}{42} = \frac{22}{42} = \boxed{\frac{11}{21}}$

(ii) "Red first" and "at least one green" happen together only on path RG $= \frac{10}{42}$:

$$P(\text{R first} \mid \text{at least one G}) = \frac{10/42}{22/42} = \boxed{\frac{5}{11}}$$

12. **[Maximum mark: 8]**

Consider two events, A and B. It is given that

$$P(A \cap B') = \frac{5+x}{24}, \qquad P(A \cap B) = \frac{7-x}{12} \qquad \text{and} \qquad P(A' \cap B) = \frac{1}{6},$$

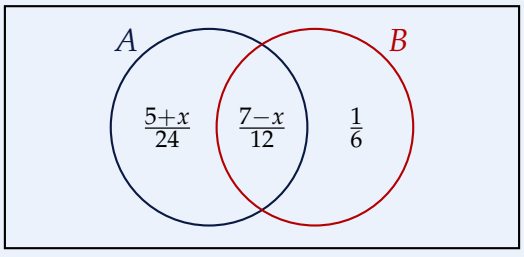
where $0 \leq x \leq 7$.

(a) Show that $P(A) = \dfrac{19-x}{24}$. [3]

It is given that A and B are independent.

(b) Find the possible values of x. [5]

Answer



(a)

$$P(A) = P(A \cap B') + P(A \cap B) = \frac{5+x}{24} + \frac{2(7-x)}{24}$$
$$= \frac{5+x+14-2x}{24} = \frac{19-x}{24} \quad \blacksquare$$

(b) $P(B) = P(A \cap B) + P(A' \cap B) = \frac{7-x}{12} + \frac{2}{12} = \frac{9-x}{12}$.
Independent, so $P(A \cap B) = P(A)\,P(B)$:

$$\frac{7-x}{12} = \frac{19-x}{24} \cdot \frac{9-x}{12} \;\Rightarrow\; 24(7-x) = (19-x)(9-x)$$

$$168 - 24x = 171 - 28x + x^2 \;\Rightarrow\; x^2 - 4x + 3 = 0 \;\Rightarrow\; (x-1)(x-3) = 0$$

$$\boxed{x = 1 \text{ or } x = 3}$$

Both are in $0 \le x \le 7$. Check that "neither" is non-negative: $1 - \frac{23-x}{24} = \frac{1+x}{24} > 0$ ✓. So both values are valid.
Common mistake: cross-multiplying wrongly. Multiply both sides by 288 $(= 24 \times 12)$ to get $24(7-x)$ on the left.

13. **[Maximum mark: 8]**

Paul and Mary play a game where they each take it in turns to roll a regular six-sided die. Paul wins the game if he rolls a number greater than 2, and Mary wins the game if she rolls an odd number. Paul rolls first.

(a) (i) Determine the probability that Paul wins the game on his first roll. [1]

(ii) Determine the probability that Paul wins the game on his second roll. [2]

(iii) Determine the probability that Paul wins the game on his third roll. [2]

(b) Show that the probability of Paul winning the game gives a geometric sequence. Hence find the sum to infinity. [3]

Answer

$P(\text{Paul wins a roll}) = P(3,4,5,6) = \frac{4}{6} = \frac{2}{3}$, so $P(\text{Paul misses}) = \frac{1}{3}$.
$P(\text{Mary wins a roll}) = P(1,3,5) = \frac{3}{6} = \frac{1}{2}$, so $P(\text{Mary misses}) = \frac{1}{2}$.
(a)(i) $P = \boxed{\frac{2}{3}}$
(ii) Paul misses, Mary misses, then Paul wins: $\frac{1}{3} \times \frac{1}{2} \times \frac{2}{3} = \boxed{\frac{1}{9}}$
(iii) Two full rounds with no winner, then Paul wins: $\left(\frac{1}{3} \times \frac{1}{2}\right)^2 \times \frac{2}{3} = \frac{1}{36} \times \frac{2}{3} = \boxed{\frac{1}{54}}$
(b) Each extra round in which both players miss has probability $\frac{1}{3} \times \frac{1}{2} = \frac{1}{6}$. So

$$P(\text{Paul wins on roll } n) = \frac{2}{3}\left(\frac{1}{6}\right)^{n-1}$$

which is a geometric sequence with $u_1 = \frac{2}{3}$ and $r = \frac{1}{6}$. Since $|r| < 1$, the sum to

infinity exists:

$$S_\infty = \frac{u_1}{1-r} = \frac{2/3}{5/6} = \frac{2}{3} \times \frac{6}{5} = \boxed{\frac{4}{5}}$$

Tip: simplify $\frac{4}{6}$ to $\frac{2}{3}$ and $\frac{6}{36}$ to $\frac{1}{6}$ first. The arithmetic is much easier.